

# The model family and the landscape

Sleeping Machines is a family of causal networks of stateful temporal programs. Content, computational time, addresses and persistent evidence jointly determine computation. A common interface supports different programs and mixtures at every level.

The envelope supplies choices; a member fixes compatible operators and contracts; the integrated target is measured on temporal/selective quality and resource behavior.

# A generative design space, from atoms to systems

Choose evidence, flow, writes, keys, reception, output and credit; then compose modules, interactions, layers, stacks and complete learning systems. Uniform width, fan-in and scheduling are optional design choices.

Review computational power and total work, represented distinctions, and useful learning at each level. Expand the earliest limiting region: downstream richness cannot recover erased evidence.

# The same three axes at every level

Computational power and required work, represented distinctions, and useful learning are separate properties, and each must survive the interfaces that compose a model.

| Level        | Power / work                                           | Representation                                        | Trainability                                             |
|--------------|--------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|
| Atoms        | Flow, delays, gates, races, sums and their ordering    | Content, order, interval, phase and repeated evidence | Continuous sensitivity plus event-boundary utility       |
| Units        | State transformation, write, emission and readout      | Retained facts, temporal modes and message bandwidth  | Content/time/state credit; conditioning and exposure     |
| Modules      | Discovery, candidate proposals, reception and commits  | Accessible evidence and which messages can meet       | Supported alternatives, selection and write utility      |
| Layers       | Mixing, shared reads, transport and timed joins        | Relations across programs, sources and values         | Coupled derivatives and changed event histories          |
| Stacks       | Depth, recurrence and optional continuation            | Hierarchical features and persistent context          | Information paths, Jacobian products and writer horizon  |
| Compositions | Neural, statistical, historical and rich local regions | Abstraction plus direct evidence at a common query    | Fusion responsibilities, retrieval and future utility    |
| Systems      | Schedule, caches, batching, optimizer and serving      | Usable state within information/resource limits       | Causal objectives, versions, update consistency and cost |

Diagnose lost information, unsupported computation and missing credit separately. Then test their coupled consequence in an integrated member, preserving the successful parent and the full resource boundary.

# Relationships to established model families

The envelope intersects existing categories and contains reference constructions for them. Its contribution is an economical integration of time, state, selection and learning; existing systems can also be extended or hybridized.

| Family                  | What can be included                                        | Additional design freedom                                  | Condition / price                                                   |
|-------------------------|-------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Recurrence / SSM        | Supported state transitions and filters                     | Addressed private programs and timed nonlinear interaction | Actual operators, retained state and stable learning                |
| Attention / Transformer | Supported attention and full reference blocks               | Persistent evidence, races and variable delivery fan-in    | Full aggregation, FFN/residual/norm/positions; all required work    |
| Sparse experts / MoE    | Supported gating and expert mixtures                        | Private stateful experts, clocks and future write utility  | Candidate discovery, mixture aggregation and trained specialization |
| Counts / retrieval      | Causal estimators and explicit historical evidence          | Shared learned abstractions and temporal/nonlocal fusion   | Correct statistics, support, access, updates and storage            |
| Adaptive memory / TTT   | Specified inner learner and causal deployment updates       | Selective temporal programs and local learning policies    | Actual objective/update, versions, retained credit and extra work   |
| Event / temporal        | Local event scheduling, timers and supported temporal logic | Small vector messages and coupled alternative-route credit | Precision, ordering, termination, traffic and measured hardware     |

A function inclusion is not a trained-quality or resource result. A restricted winner-only member has less interaction support than a full-aggregation member. Broader operator choices provide a route to matching needed functions without promising that present native units already implement every reference.

## Units combine content, timing and persistent evidence

The native receiver below illustrates one branch. It combines a stored key, input-conditioned temporal proposal, small residual value and computational clock. Keys choose access; values carry content; the selected write changes future evidence.

A unit is a product of choices: statistical fields, neural vectors, protected facts or fast parameters; identity, decay, rotation or richer local maps; first, multi-message, window or silence reception. Their integration changes the function and the credit needed to learn it.

## Modules, interactions, layers and stacks

Candidate discovery, winner selection, reception and continuation are distinct decisions. Heads expose complementary evidence only through declared mixing, joins or shared reads. The diagram is the native branch, not the whole family definition.

Depth creates hierarchical temporal functions when information and credit survive interfaces. Persistent facts, accessible facts and trainable writers are separate capacities. Shared maps do not automatically share private state across sources.

# One family across dense and sparse worlds

Dense frames, language, queries and irregular events can feed explicit shared information paths. Input density, computation density, arrival cadence and physical clocking are independent choices.

A rich local island can reproduce supported attention or other reference operators while surrounding regions remain selective. Full participation needs actual aggregation; synchrony needs snapshots/barriers. Joint multimodal training remains an integration ambition. No global periodic tick is required; physical clockless energy is unmeasured.

## The same structure can learn, batch and adapt

Representation, timing, routes and writes form one coupled credit problem. Factual derivatives teach a realized history; useful alternative returns teach changed routes, reception and later state consequences. The successful local value teacher does not cover full future-write utility.

An earlier neural family pilot improved causal online prediction at 10.736 times total processing work. It is scoped evidence, not current native-backbone TTT or drift retention. State/statistical adaptation is distinct from gradient TTT. Local, streamed and batched updates need ownership and versions. General morphing and fully asynchronous hardware training remain open; learned event routes already exist.



# A precise core with contrasting complete members

A member declares the following twelve semantic fields. A construction grammar permits serial, parallel, recurrent, memory-linked, scheduled and policy-controlled regions, subject to compatibility and termination.

| Field             | Required contract                                                              |
|-------------------|--------------------------------------------------------------------------------|
| E / Evidence      | Observation schema, provenance, time/address adapters and admissible histories |
| O / Operators     | Actual maps, domains, precision and rounding for numerical claims              |
| G / Composition   | Units, connections, shared paths and structural-action space                   |
| S / State         | Facts, timing, queues, caches, randomness, ownership and resets                |
| I / Interfaces    | Content/address/time types, conversions and state access                       |
| P / Participation | Discovery, keys, proposals, deliveries, writes and alternative support         |
| T / Schedule      | Order, delays, ties, deadlines, joins, cancellation and barriers               |
| Q / Queries       | Observed cutoff, accessible snapshot, completion and readout                   |
| L / Objectives    | Task/self-supervision, labels, scoring order and resource penalties            |
| U / Learner       | Credit, horizon, optimizer/update ownership and versions                       |
| B / Bounds        | Finite execution, capacities, admission and overflow policy                    |
| X / Execution     | Reference/deployment equivalence, measurement and evidence status              |

C: causal binary counts, addressed increments and a full prediction distribution; no gradient TTT. R: two private temporal programs, hard race and a local online value teacher; no deep or full writer credit. H: shared sparse reception, two-record dense query and exact finite-option episode TTT. These complete illustrative specifications share one template; they are not new fitted models or benchmark claims.

## Composition and adaptive growth have separate contracts

Local capability reaches the system only through compatible information, state, time, query, credit and resource interfaces. Serial, parallel, recurrent and hybrid compositions expose different evidence and dependencies.

Forward preservation checks future state and emissions, including pending deadlines. Learning preservation additionally checks credit and parameter/optimizer migration. Resource refinement checks actual work, storage, traffic and latency. Equal current predictions do not establish any of these stronger claims. Intended learned deviations remain permitted.

# Adaptive computation keeps the task and query contract fixed

Learned routes, delays, reception and stopping change internal execution. Ordinary quality comparisons still score the same declared targets. If a branch is unfinished, specify the accessible snapshot and fallback or mark the run invalid. Internal silence need not mean the system declines to answer.

| Choice                     | Query / learning obligation                                           | Full resource boundary                                   |
|----------------------------|-----------------------------------------------------------------------|----------------------------------------------------------|
| Slow route / deadline      | Name the clock; use the declared snapshot/fallback; retain the target | Attempted work, pending state and permitted waiting      |
| Stopping / optional depth  | Score the actual stopped readout for every declared target            | Discovery, rejected proposals and executed continuations |
| Silence reception          | Explain how queries read a pending or empty receiver                  | Timers, cancellation, retained state and readout         |
| Online / structural update | Predict before that outcome; preserve feedback and version rules      | Credit, replay, migration and optimizer                  |
| Random execution           | Declare repeats; pair quality and cost from the same realization      | Distributions/tails when budgets or deadlines matter     |

A simple warning: answering only the easy half can report zero conditional loss at 50% coverage, while full-workload loss with a unit fallback penalty remains 0.5. Selective prediction needs its own coverage/risk protocol. This is a semantic example, not a discovered error in saved language scores.

For finite choices, route credit is  $\pi_i$  times  $(Q_i \text{ minus mean } Q)$ , where  $Q_i$  is the full conditional task/resource return. Immediate value utility omits unmodeled future writes, deadlines and work. More alternatives do not correct a wrong utility definition. Modeled time is not measured hardware latency.

## A shared substrate opens input, communication and embodied opportunities

The opportunity register describes six application/mechanism directions with precedents, status and first proof conditions. The diagram illustrates a proposed event-native codec: learned memory and send/wait policies serve fixed reconstruction or task queries.

Irregular inputs can preserve time and recruit local programs. Quantized asynchronous outputs could allocate communication where useful. Count actual content, address, timing, framing and reset bits, plus both endpoints' work; variable message count is not itself a bitrate result.

Meta-learning offers fast-state/slow-rule mechanisms for retention, retrieval, writing and scheduling. Shared sensor/action/language paths could let embodied experience teach cognitive representations and reasoning guide action. Measure each transfer direction on held-out tasks with matched data/work and disconnected-path controls.

Existing variable-rate codecs, irregular-stream models and vision-language-action systems are relevant precedents. Our opportunity is their useful temporal/selective integration. No project codec, dexterity-transfer, AGI, customer or hardware-energy result follows from this chapter. Existing integrated queues retain priority.

# Evidence supports the family; integration sets the agenda

Completed native results support content-bearing temporal computation, useful route credit and capacity beyond selected writes. The rows below share T256 and the same restricted 1M test region; each is single-seed. The 90M row changes data and width.

| Native member                | Fit M  | BPC    | Fit TFLOPs | MFLOPs/target |
|------------------------------|--------|--------|------------|---------------|
| P32 / U2, timing-only credit | 9.994  | 2.5064 | 7.221      | 0.722         |
| P32 / U2, credit             | 9.994  | 2.3715 | 7.243      | 0.725         |
| P32 / U4, credit             | 9.994  | 2.3452 | 11.950     | 1.196         |
| P64 / U2, 90M data           | 89.997 | 1.8573 | 241.219    | 2.680         |

Saved arithmetic plus unit-counted special functions; eager full-step estimates. Whole-fit and per-fitting-target columns have consistent units across rows. Evaluation, setup, traffic and energy are additional.

Value choice credit improves the matched 10M D4/P32/U2 result by 0.134895 BPC for about 0.30% extra estimated fitting work. U2→U4 adds useful capacity with the same eight writes, but increases fitting work and untied parameters. These support scoped mechanisms; they are not modern frontier or public full-test wins.

Next integrated proof points: complete future-write utility and useful horizons; discovery/reception/silence; trained sparse-state and quality parity; matched repeated controls and measured traffic/energy. Native TTT, automatic structural design and shared-world integration require their own evidence.

The investment thesis is a reusable model/runtime/compute substrate from datacenters to mobile and robotics. The family definition makes that breadth concrete; commercial advantage remains an empirical quality/resource and adoption question.